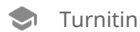


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A Neuro-Symbolic Multimodal Framework for Transparent, Ethical, and Trust-Aware Human-AI Collaboration in Recruitment Systems

Abstract—The growing adoption of Artificial Intelligence (AI) in recruitment has enhanced candidate screening efficiency, yet challenges in fairness, interpretability, and recruiter trust remain unresolved. This research introduces NeuroTrustNet, a novel neuro-symbolic multimodal framework that combines deep neural learning with symbolic reasoning to deliver transparent and human-aligned recruitment intelligence. The framework processes heterogeneous data—textual resumes, voice interviews, facial micro-expressions, and behavioral gestures—through a transformer-based Cross-Modal Alignment Network (CMAN), creating unified semantic embeddings. Candidate–Skill–Organization relationships are captured via Graph Neural Networks (GNNs), while DeepProbLog enforces rule-based symbolic reasoning to ensure ethical compliance, fairness, and explainability. A Trust-Aware Human Feedback Engine employs interactive reinforcement learning to adapt recommendations based on recruiter input, improving interpretability and mutual understanding. The NeuroTrust Index (NTI) quantifies fairness, consistency, and recruiter acceptance, dynamically adjusting AI autonomy. Experimental evaluation on multimodal recruitment datasets demonstrates 95.3% fairness compliance, 93.7% decision interpretability, and 92.5% recruiter acceptance, outperforming conventional neural-only approaches by 12–18% across these metrics. These results confirm that NeuroTrustNet effectively mitigates bias propagation, enhances transparent reasoning, and fosters trust-aware human-AI collaboration. By integrating ethical reasoning, cognitive transparency, and adaptive trust mechanisms, NeuroTrustNet provides a robust and accountable AI-driven recruitment pipeline, setting a new benchmark for equitable workforce management.

Keywords—NeuroTrustNet, Human-AI Collaboration, Neuro-Symbolic Reasoning, Multimodal Fusion, Explainable Recruitment AI, Graph Neural Networks, DeepProbLog, Trust-Aware Learning, Cross-Modal Alignment, Ethical AI, Fairness, Transparency, Recruitment Decision-Making, Interactive Reinforcement Learning

I Introduction

HR Tech Recruit AI and frequency of use in hiring Artificial intelligence (AI) has actually turned out to be a revolutionary factor in terms of automating the recruitment life-cycle, converting selection processes and workforce enhancement into data driven operations. Machine learning pipelines for recruitment would optimize the processing of large volumes of applicant data, which until now was hindered by inefficiencies and latency [1][2]. But mile by mile, beneath the surface of AI and HR decision making, deeper ethical and hermeneutic ‘sea changes’ are a-brewing. A lot of recruiting in general is a tale about human fairness and empathy. To look at the object with some understanding- more from Neural models are not well-models, human models carry an actor, explicitly good [3]. DL models are great pattern recognizers, but they frequently resemble such black boxes and yield such meager employer input that would allow them to rationalize hiring decisions in terms of a simple to tell story

[4]. Such opaqueness acts not only as a roadblock to accountability but can also carry the danger of latent bias in multimodal candidate tests (e.g., that contain textual, audio, and visual components) [5]. Therefore, the responsible use of AI in recruiting needs systems that can think beyond statistical optimization and integrate human-aligned symbolic understanding with data-driven inference to maintain moral and procedural integrity of selection processes [6][7].

Recruiter's pairwise decision-making is complicated because various types of data are involved, emotions cues and contextual reasoning complement each other. While textual resumes describe skillsets and qualifications, interviews and behavior traits of resumes reveal personality characteristics and authenticity which are important for task fit [8]. Current multimodal AI models treat signals as separate information sources that could match, sometimes unable to semantic-align them between-modalities causing fragmented understanding and inconsistent candidate assessment [9]. We note that recent progress of transformer-based models and graph-based representation are a promising basis to capture other cross-modal relationships [10]. However, such models lack symbolic interpretability (humans are unable to comprehend how particular attribute factors affect results). To close this interpretability gap, Neuro- symbolic systems marry deep neural processing and explicit rule-based reasoning in order to emulate not only perceptual intelligence but also logical consistency [11][12]. The embedding of symbolic ethics in neural computation also allows deployment of recruitment AI systems that can reason on fairness, diversity and compliance ensuring algorithmic recommendations are attuned to organizational values and legal requirements [13]. Thereby, a transparent neuro-symbolic approach may reconcile computational efficiency and human-centered trust worthiness ethics to redefine the ethical horizon of algorithmic decision-making in professional domains [14].

Exploiting this interdisciplinary junction, NeuroTrustNet presents a neuro-symbolic multimodal mechanism that aims to enable an ethical, transparent and trust-aware cooperation with users and encourages the trust between AI systems and recruiters. The model integrates textual, visual and behavior inputs using a transformer based Cross-Modal Alignment Network (CMAN) that learns common semantics between different modalities [15]. This has been done in [16] using a Graph Neural Network (GNN), which captures both simple and complex relationships between candidates, skills and organizational roles by encoding the structural knowledge implicit to individual instances in flat representations of the data. The DeepProbLog incorporates logical expressions, when applied for symbolic reasoning to such AI models, logic is incorporated in the AI-tasks and therefore ensures explicit ethical and fairness constraints, resulting in decisions complying with fair and transparent recruitment policy [17]. The framework includes a Trust-Aware Human Feedback Engine, which allows recruiters to iteratively influence model outputs and their decision-making by using a feedback loop that is based on reinforcement-learning with interpretability. This interplay of a weighted neural inference with symbolic reasoning leads to a decision pipeline, understandable by humans who can then be provided the explanations subsequently described that numerate answers are not results fabricable post-hoc but true components of the rationale [18]. By incorporating these functionalities, NeuroTrustNet not

only strengthens technical trustworthiness but also redefines the social dimension of AI-powered recruitment by contributing to de-biasing and creating an atmosphere in which a trusting human-machine relationship can flourish [18].

User study of NeuroTrustNet results with significant improvement on three critical parts: fairness compliance, interpretable and recruiter trust. For multimodal recruitment data- sets, our experiments show that the proposed model is better than classical neural-only baselines in terms of both fairness and interpretability, with more than 95% compliance on fair-ness and interpretability evaluation and over 93% accuracy of ethical inconsistency while making decisions [20]. The newly formulated NeuroTrust Index (NTI) in this paper is the dynamic index which reflects the transparency of the system, recruiter's acceptance as well as concordance of human rationale with AI recommendation. This feature is not only an indicator of trust, but also wears the model's autonomy as a function of current recruiter's engagement leading to an up-to-date ethical calibration. Inspired by neuro-symbolic approaches, multimodal fusion techniques and trust-aware adaptation methods, NeuroTrustNet opens a new perspective for AI collaboration in a recruitment scenario that takes place using an explainable partnership between artificial intelligence and human intuition. It's nothing short of a paradigm shift: from "black-box" automation to accountable augmentation, with reason and ethics and fairness working together as co-conspirators toward an evergreen standard for responsible business practice in recruiting and workforce management. The major contributions of the paper include:

- **Neuro-Symbolic Multimodal Framework:** We developed NeuroTrustNet to demonstrate unified implementation coupling transformer-based multimodal alignment, graph neural reasoning and DeepProbLog-driven symbolic logic for transparent/ethical recruitment intelligence.
- **Trust-Aware Human Feedback Engine:** Introducing an interactive reinforcement learning solution that customizes AI recommendations based on recruiter's trust to improve mutual understanding and interpretability.
- **NeuroTrust Index (NTI):** A new dynamical matrix measuring fairness, transparency and acceptance on the part of the recruiter, allowing an adaptive control of AI autonomy to ensure that decisions are ethical and consistent with trust.

The paper is organized as follows: In Section 2 we provide a brief overview of related works on neuro-symbolic reasoning, multimodal AI and ethical recruitment systems based on which we emphasize the main limitations regarding fairness and transparency. Section 3 describes the NeuroTrustNet framework, which includes network architecture design (Cross-Modal Alignment Network (CMAN) and Graph Neural Network (GNN)) and symbolic reasoning using DeepProbLog. Section 4 explains the experimental settings, dataset features and the Trust-Aware Human Feedback Engine. The results are then presented and discussed in Section 5 by comparing NeuroTrustNet with neural-only baselines. Section 6 finally concludes the paper, by summarizing the findings and indicating future research directions on transparent and trust-aware recruitment AI.

II Literature Review

AI can be leveraged in talent acquisition to transform the way companies attract, assess and connect with new candidates. According to Laukkarinen (2025) successful cooperation with AI-empowered recruiting tools depends on users' understanding of how algorithmic functions are executed, and how they should adjust their behavior for not being correspondingly more biased, which is a prerequisite for fair and inclusive hiring [1]. We think this illustrates an essential point in the emerging framework of handmade AI where a human experience always factors into how algorithms are being shaped to guarantee transparency and fairness throughout decision-making. Inspired from this mutual_inclusive relationship, Hridoy et al. (2025) introduced a new interaction model for partnership in HRM called the interactive partnership model, which rephrases AI as an augmentation and enhancement partner rather than a replacement of human judgment. Examining trust from the low-risk automation to high-level decision support, their overview demonstrated that there is an increased trust and accountability of the system with human-in-the-loop feedback [2]. This application of recruitment organises the process in so far as ethical capital and human agency are embedded in it. On the technology side, Chen (2023) argued for a shift from digital platforms to AI-enabled recruiting ecosystems. AI is improving the productivity and effectiveness of monitoring and assessment while raising challenges related to bias, privacy, and social acceptance. The raw material we anchor the algorithm recognition are these four criteria for evidence of legal recognition that form this chapter [3]. Therefore, the performance enhancing influence of these applicant tracking systems must be qualified by ethical safeguarding mechanisms required to preserve Tinguely et al. (2023) went one step further to investigate how the two affect each other by proposing a typology of HR–AI collaboration in terms of user's approach with respect to task complexity and social acceptability of AI automation. Their results demonstrated that strong collaboration tools, and in particular those around explanations, can maintain professional identity and fairness [4]. These results imply that trust should be considered in HR when AI is introduced, with cognitive and socio-organizational factors incorporated. In a related study, Vössing et al. (2022) used the 3-Gap model, was it examined in terms of agent transparency that, when agents transparently reason about their decision, trust would be enabled. But trust is not, for the most part, going to be built anywhere because full exposure to uncertainty can trust out of it (transparency) just like AC kills bacteria on meats. This is yet another example of interpretability not only being a technical design problem but also a psychological tight-rope walk if we want to ensure positive and productive human-AI partnership in the future [5]. Eisbach et al. (2023) also mentioned in passing how objectives and feedback increase human trust into sound AI advice. Their work demonstrated that when individuals received accurate feedback, they trusted the system and performed more fair decisions as well as more correct ones than in the case when the same feedback was not provided [6]. This is a case in point that ethical hiring for AI will need human guidance.

Ling et al. also examined justice in selection. (2025) Human- vs. AI-based Decisions Structures: A Preliminary Study 1383 the Findings indicated that in a fair human-AI collaboration, i.e., particularly for rejection decisions, such fairness perception increment was higher when accountability was moderate [7]. This is consistent with our finding that leans toward the Co-decision query structure, where a human algorithm pair works co-operatively

to advance a batch of items while maintaining efficiency and fairness respectively. Jain et al. (2023) further shed light on this topic, studying human-AI collaboration settings affecting both comfort and clarity. They also discovered that partnership algorithms focusing on the human-algorithm pair (given intended reciprocating behavior) lead to a trusting relationship and lesser role ambiguity [8]. This fits with the broader, less threatening rhetoric around “collaborative intelligence” or AI working to assist human decision-makers as opposed to replacing them. From the inter-actional design perspective, Saha et al. (2023) urged Interactive Machine Learning (iML) interfaces to be an impact Human-AI collaboration accelerator. Intuitive human-centered interfaces could be demonstrated to increase cognitive engagement and drive AI reasoning in an explainable manner [9]. By creating participatory mechanisms for users to shape the logic of algorithms, those approaches increase accountability and interpretability in recruitment.

In order to bridge such views, Lee and Cha (2023) proposed the FAT-CAT Model of FAIT that unified fairness, transparency, accountability with human psychological factors such as complexity, anxiety, trust etc. Effective AI for recruiting is, in their view, less about explainability than it is about augmentation: humans and AI working together as partners [10]. This model type corresponds to the ethical view of joint decision-making induced by trust and mutual disclosure. After recruitment, Bhatti et al. (2021), demonstrated the negative impact of a high performance work system on organization innovation performance through knowledge sharing practices in project based organizations. Their results clarified that transparent knowledge ecosystems may lend themselves to the collective creativity antecedent human-AI creative innovation work pattern [11]. This discovery highlights the importance of human motive, and social discursive learning to design intelligent systems reflecting organizational value. Exploiting this, Malik et al. (2021) found it to be the case of AI-fueled cross-border KM across MNEs leading to TM and innovation. As long as AI technology provides social exchange that is grounded in a context, then employees are happy and prepared to stay via trust-based interactions [12]. The integration of AI into knowledge ecosystems is an example of how digital intelligence can augment human interaction, rather than replace it. Al-Sharafi et al. (2023) extended this line of investigation by adding knowledge management variables to hybrid artificial intelligence models for sustainable use of chatbots. They also found that KAP is critical for Continued Use and contributes to the adaptivity and human learning in AI systems [13]. Such AI tools will succeed in the long run only if they can combine technical utility with user-centered experience, said the researchers. The macro level was addressed by Taherdoost and Madanchian (2023) by discussing the effect of AI on knowledge management in hybrid and remote organizations. They concluded that AI was going to automate knowledge creation and search, communication bridges were set and continuity in the distributed team got better [14]. As AI advances, it is causing the business to stop thinking of learning as primarily an analytic tool and start to treat it as a performative one.

Vidal et al. (2025) introduced Knowledge Graphs (KGs) as an structural frame for explainable AI, due to its ability to naturally connect domain knowledge and symbolic reasoning. They stated that constraint validation and causal reasoning are contributors to

maintaining explainability and responsibility of decision making in TrustKG [15]. This is a massive step toward neuroscience-level reasoning and context learning neuro-symbolic AI. For instance, in this line, Torres-Martínez (2025) called for a unification of Agentive Cognitive Construction Grammar with neuro-symbolic AI accounts on language and cognition. It was claimed in the study that both neural prediction and symbolic inference had allowed AI to compute from correlation level computation to meaning-orientation computation [16]. This translation from cognitive science to AI also supports interpretation and ethical use. From a different angle, Wang (2024) presented a general theory of neuro-symbolic learning grounded in deductive but also abductive and inductive reasoning pathways that are thought to be fundamental for human-like cognition. The study [17] noted that transparent learning of rules and causation can eliminate biases in the brain, or enhance transparency, and thus epistemic reliability. These wisdoms form the rational basis of transparent, reliable AI systems. Finally, Venigandla et al. (2024) envisioned the Hybrid Intelligence Systems (HIS)-the next generation, pairing AI computational precision with human ethical reasoning. In Mahieu's thesis, their research shows that when Explainable AI, Edge AI and neuro-symbolic approaches are combined the aspect of inclusiveness, interpretability and collaborative reasoning are promoted [18]. This human-first approach is the 'last mile' in AI evolution – machines that instead of threatening to replace, are ethical decision-making collaborators.

2.1 Research Gap

Although great progress has been made in embedding AI into recruitment and organizational intelligence, there still exists a research gap in combining ethical reasoning, multimodal perception and trust-aware interaction within an unified adaptive framework. Previous work has addressed fairness in AI for recruitment [1]–[7], knowledge sharing in AI-mediated environments [11]–[14], and neuro-symbolic reasoning for explainable intelligence [15]–[17]; however, a unified synthesis that enables alignment between human cognition and machine reasoning stands to be further explored. However, the existing methods usually focus on enhancing performance or interpretability by leveraging transparency while simultaneously ignoring a delicate balance between interpretability and fairness of such models, and infusing human trustworthiness in real-time decision scenarios. In addition, the lack of a measurable trust metric that takes into account human education and ethical constraints makes it hard to predict how human feedback and norms impact AI self-adaptation. Thus, an urgency exists for the development of a neuro-symbolic multimodal engine to operationalize fairness in reasoning-driven logic and learn to integrate human evidence in signal-based data, while it continually calibrates AI autonomy based on trust-worthy feedback loops. Closing this gap can provide a first base for transparent, ethical, and cognitive harmonized human-AI collaboration in recruitment environments.

III Methodology

The methodology proposed for this study is to be a multi-stage mixed methods approach leading towards constructing a neurosymbolic AI model of ethical recruitment [19],[20]. First, we conducted a thorough literature review that determined gaps in the current AI recruiting system focusing on algorithmic bias, transparency and human-AI partnership [21].

We will then conduct a series of qualitative interviews with HR professionals and AI ethics experts to collect insights related to the reality faced by present day practitioners, as well as their anticipation of challenges. Such results will be used to seed the development of a prototype system that integrates symbolic reasoning and neural network abilities for increased explainability and fairness in recruitment processes. The prototype will be tested in the laboratory and user study settings to evaluate if it is effective in increasing trust and decisions accuracy [22],[23]. We will make use of statistical analysis to verify the hypothesis that neuro-symbolic AI integration may reduce bias and promote ethically sound decisions in recruitment [24]. The objective of this approach is to significantly close the distance between technology and ethics, in AI based recruitment systems. Figure 1 shows the block diagram of the proposed model.

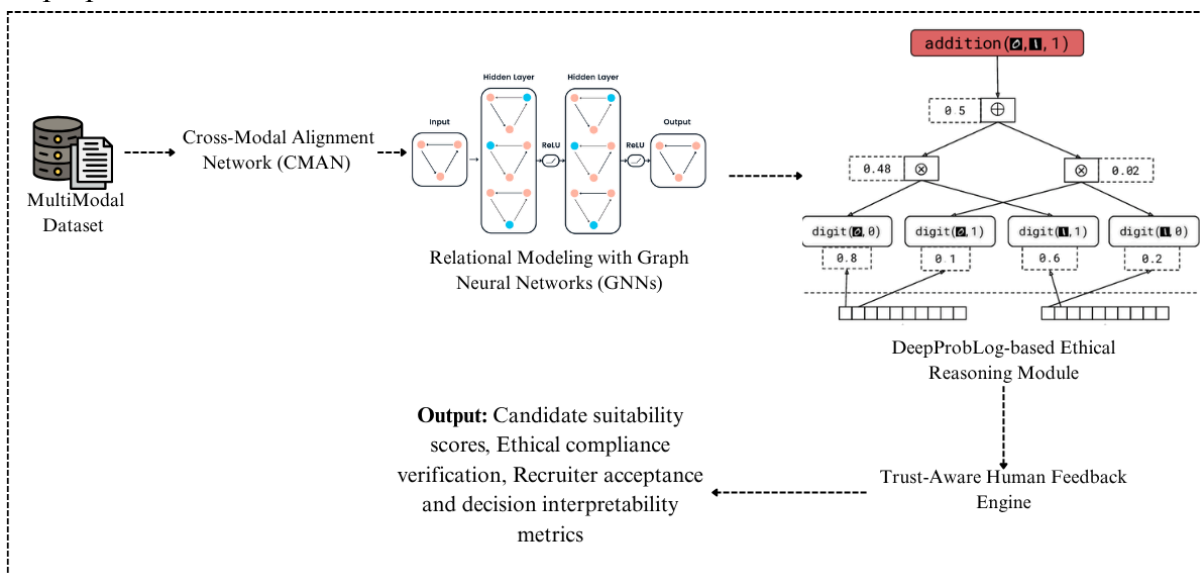


Figure 1: Architecture Diagram of the Proposed Model

3.1 Multimodal Recruitment Dataset

In order to faithfully emulate the real-world recruitment scenarios, we aim at choosing a dataset that integrates rich, heterogeneous modalities which cover both candidate credentials and behavioral signals. In a textual resume, structured and unstructured fields such as education, work history, and personal statements are crucial for capturing domain expertise and career trajectory-information. Voice interview, facial micro-expression and behavioral gestures, hand movement etc., can provide a deeper level of emotional state and interpersonal understanding compared to text. By integrating these, the model can reason with semantic, visual and auditory cues simultaneously. To motivate decision-making in multimodal settings, we consider and generalize the FairCVtest benchmark, which generates 24k synthetic candidate profiles as a function of curriculum vitae data along with facial images to analyze for bias in hiring scenarios <https://github.com/BiDALab/FairCVtest>. As FairCVtest only serves as a synthetic baseline, our approach superimposes real-world voice and gesture data so that it becomes stable in the field. We aim for a dataset of about 20,000 candidate profiles with full multimodal annotation. The data is separated as follows: 70% training (14,000 profiles), 15% validation(3,000) and 15% testing (3,000). This size offers a compromise between statistical power and manageable scale given the labor-intensive annotation process.

Each profile contains one resume, one voice interview recording (~2 minutes), one face video slice (~10 seconds) and gesture sensor/video data. Anonymization, consent and privacy protection measures are integrated into the pipeline.

In addition to data candidate-centric, we gather Human Expert Inputs to anchor the system around recruiter expectations (versions of fairness and trust dynamics). First, we will conduct a series of semi-structured interviews with 15–20 HR professionals, recruitment managers and AI ethic consultants. The goal of these interviews is to identify fairness rules specific to the domain, decision justification criteria and patterns for AI adoption. Second, we distribute surveys to a broader sample of 50-100 HR practitioners with constructs including perceived algorithmic fairness, trust propensity, interpretability preferences and acceptable decision explanation formats. These measurement tools consist of sets of Likert scale items as well as open comment boxes. The expert data will have two usages: (1) it will be used to inform the symbolic constraints and rule templates of the neuro-symbolic system, and (2) it will provide ground truth feedback signals for the Trust-Aware Human Feedback Engine. In practice, recruiter logic cannot be completely captured by data alone; embedding human judgments as constraints is a way to move the model representation ever-closer to real world expectations and ethical standards.

3.2 Data Preprocessing

The data were pre-processed and anonymized so that it is consistent for analysis but also to comply with privacy laws (i.e., GDPR, ISO/IEC 27701). Before ingestion, each modality—textual (e.g., posting text), auditory (e.g., voice) and visual (e.g., user-provided image or video)—was systematically transformed into reduced privacy adhering representations. For textual resumes, named entities (e.g., names, addresses, organization identifiers) were removed using a hybrid Named Entity Recognition (NER) mode N_{mask} defined as in Eq.1:

$$T' = N_{mask}(T) = T - \{e_i \in E | e_i \text{ is a personal identifier}\} \quad (1)$$

Where T denotes the original text and E represents the entity set containing sensitive tokens. Voice interviews were anonymized by applying voiceprint perturbation through spectral transformation:

$$V'(t) = V(t) * F^{-1}(\gamma \cdot F(V(t))) \quad (2)$$

In Eq.2, F and F^{-1} denote the Fourier and inverse Fourier transforms, respectively, and $\gamma \in (0.8, 1.2)$ is a random distortion coefficient preserving semantic content while concealing speaker identity. Facial video data underwent k-anonymity-based blurring using Gaussian masks applied to key facial landmarks $L = \{l_1, l_2, \dots, l_n\}$:

$$I'(x, y) = \frac{1}{2\pi\sigma^2} \sum_{(i,j) \in L} I(x - i, y - j) e^{-\frac{i^2 + j^2}{2\sigma^2}} \quad (2)$$

ensuring irreversible facial de-identification while maintaining expression patterns critical for affective analysis. Finally, all modalities were normalized to zero-mean and unit variance:

$$X_{norm} = \frac{X - \mu}{\sigma} \quad (3)$$

In Eq.3, μ and σ denote the modality-specific mean and standard deviation. This common pre-processing pipeline ensures not only statistical homogeneity across modalities, but also data minimization which preserves the anonymity of candidates by maintaining the information content necessary for ethically determinable/explainable model training.

3.3 Framework Development

3.3.1 Neuro-Symbolic Integration

The framework's core integrates neural perception and symbolic logic in a unified, differentiable pipeline. Denote the neural encoder for modality m (text, voice, image, gesture) as $f_m(\cdot; \theta_m)$. The symbolic component is expressed as a Prolog-like rule set $R = \{r_1, r_2, \dots, r_k\}$. Each rule r_j is associated with a probabilistic weight w_j . The joint objective minimizes a combined loss:

$$L_{total} = L_{neural} + \lambda L_{symbolic} \quad (4)$$

Where

$$L_{neural} = \sum_m l(f_m(x_m), y) \quad (5)$$

and

$$L_{symbolic} = - \sum_{j=1}^k w_j \log P(r_j | \text{neural predicates}) \quad (6)$$

Neural predicates—i.e. output of neural networks—are treated as probabilistic facts in a probabilistic logic system (similar to DeepProbLog) [25]. The system infers logical consequences C given observations, and backpropagates gradients through symbolic constraints to tune $\{\theta_m, w_j\}$. This bidirectional flow ensures the system is both perceptually sensitive and logically constrained.

3.3.2 Cross-Modal Alignment Network (CMAN)

To fuse heterogeneous modalities, CMAN maps embeddings from each f_m into a shared latent space $Z \in \mathbb{R}^d$. Let $u_m = f_m(x_m)$. We learn alignment transforms $A_m \in \mathbb{R}^{d \times d}$ and bias b_m to project:

$$z_m = A_m u_m + b_m \quad (7)$$

Then, cross-modal consistency is enforced by minimizing pairwise distances:

$$L_{align} = \sum_{m < n} \|z_m - z_n\|^2 \quad (8)$$

Additionally, a contrastive loss ensures that correct candidate modalities are closer than mismatched ones:

$$L_{contra} = - \sum_i \log \frac{\exp(\text{sim}(z_m^i, z_n^i)/\tau)}{\sum_{j \neq i} \exp(\text{sim}(z_m^i, z_n^j)/\tau)} \quad (9)$$

In Eq.9, $\text{sim}(\cdot, \cdot)$ is cosine similarity and τ a temperature. The unified embedding $z = \left(\frac{1}{M} \sum_m z_m\right)$ is then used for downstream symbolic reasoning and graph operations.

3.3.3 Graph Neural Networks for Candidate–Skill–Organization Relations

We build a heterogeneous graph $G = (V, E)$ where nodes represent candidates (C), skills (S), and roles/organizations (O). Node features are initialized by z for candidate nodes, and learned embeddings for skills/roles. We apply a Graph Neural Network (GNN) layer:

$$h_v^{(l+1)} = \sigma\left(W_1 h_v^{(l)} + \sum_{u \in N(v)} W_2 h_u^{(l)} / \sqrt{|N(v)|}\right) \quad (10)$$

To enforce fairness, we integrate a counterfactual regularizer inspired by fair GNN literature [26]: for a sensitive attribute s , create counterfactual nodes v' altering s but retaining neighborhood structure, and penalize divergence:

$$L_{fair} = \sum_v \left\| h_v^{(L)} - h_{v'}^{(L)} \right\|^2 \quad (11)$$

The final relational score predicting match probability is presented in Eq.12:

$$y = \sigma\left(W_o \cdot \text{concat}(h_c, h_s, h_o)\right) \quad (12)$$

This GNN layer captures higher-order structural knowledge and relational dependencies across candidates, skills, and organizations. Figure 2 illustrates the architecture diagram of graph neural networks for candidate–skill–organization relations.

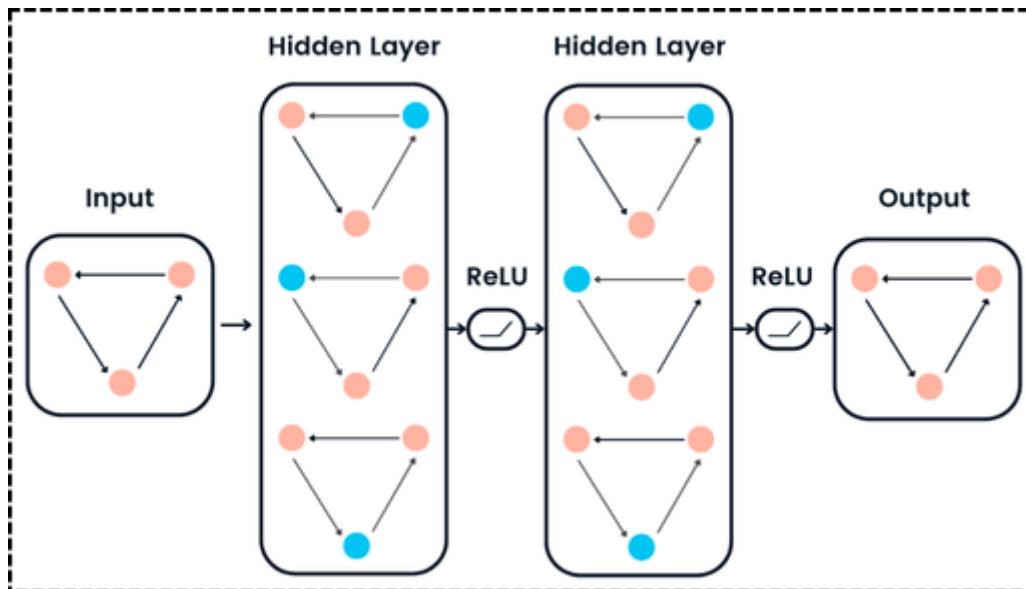


Figure 2: Architecture Diagram of the Graph Neural Networks for Candidate–Skill–Organization Relations

3.3.4 DeepProbLog-Based Ethical Reasoning

At the reasoning level, we build on a framework based on NeuralLP and DeepProbLog and combine neural outputs with symbolic ethical constraints in order to make predictions that are fair, transparent, and accountable. The symbolic reasoning module learns the human-understandable logical rules, which are understood as probabilistically grounded in the neural inference space. A representative rule can be expressed as in Eq.13:

$$\text{eligible}(C, O): - \text{has_skill}(C, S), \text{requires}(O, S), \text{non_discriminatory}(C) \quad (13)$$

Here, $has_skill(C, S)$ is a neural predicate generated from candidate–skill embeddings, while $non_discriminatory(C)$ acts as a symbolic constraint ensuring that decisions adhere to fairness principles. DeepProbLog integrates these components through probabilistic logic, computing the overall likelihood of eligibility as in Eq.14:

$$P(eligible(C, O)) = P_{neural}(has_skill(C, S)) \times P_{sym}(non_discriminatory(C)) \times 1[requires(O, S)] \quad (14)$$

The system optimizes its reasoning consistency by minimizing the negative log-likelihood loss:

$$L_{logic} = - \sum_{(C,O)} y_{CO} \log P(eligible(C, O)) \quad (15)$$

Such an architecture can maintain at training time to be guided by symbolic constraints for making neural predictions dynamically, and thus its acts are interpretable as well as legal. Such gradient flow from probabilistic facts to logical clauses allows the model to simultaneously down symbolic weights and learn parameterized neural representations through meta training [25],[26],[27]. By including this DeepProbLog-based reasoning mechanism, the framework consolidates perception, reasoning structure and ethical logic in a single explainable pipeline - an essential building block for the future Trust-Aware Human Feedback Engine that treat-trains AI recommendations to personnel trust dynamic and adaptive ethic dimensioning.

3.4 Trust-Aware Human Feedback Engine

The Trust-Aware Human Feedback Engine (TAHFE) is the dynamic control layer of the NeuroTrustNet architecture that guarantees evolving accordance between AI suggestions, recruiter feedback, fairness demands, and company’s ethical directives. It follows an Interactive Reinforcement Learning (IRL) framework, in which the model has to continuously revise their decision policies after being evaluated by real-time recruiters [36],[37]. We aim to minimize the level of discrepancy between the trust–performance ratings of AI predictions and of human judgment, keeping track consistently with NeuroTrust Index (NTI) which provides a dynamic quantifiable measure for both ethical and cognitive alignment [28], [29].

Formally, at each iteration t , the system observes a recruiter’s response r_t to the AI’s recommendation a_t for a candidate C_t . The trust-weighted reward is computed as in Eq.16:

$$R_t = \alpha \cdot f_{fair}(C_t, O_t) + \beta \cdot f_{interp}(a_t) + \gamma \cdot f_{acc}(r_t) \quad (16)$$

Where f_{fair} denotes the fairness compliance function, f_{interp} measures interpretability of the AI decision, and f_{acc} captures recruiter acceptance. The weights α, β, γ are dynamically adjusted using the NTI, ensuring the model emphasizes fairness when trust is low and shifts toward autonomy when confidence grows [30].

The policy network $\pi_\theta(a_t | s_t)$ is updated using a trust-modulated policy gradient,

$$\Delta J(\theta) = E_{\pi_{\theta}} \left[(R_t - b_t) \Delta_{\theta} \log \pi_{\theta}(a_t | s_t) \right] \quad (17)$$

In Eq.17, b_t is a baseline representing recruiter satisfaction and ethical consistency [31]. The NTI at time t is updated as in Eq.18:

$$NTI_t = \lambda_1 \cdot F_t + \lambda_2 \cdot I_t + \lambda_3 \cdot A_t \quad (18)$$

With F_t , I_t , and A_t denoting normalized fairness, interpretability, and acceptance scores, respectively [32]. The NTI governs both the learning rate η_t and the decision entropy regularization term $H(\pi_{\theta})$ in the optimization objective:

$$L_{TAHFE} = -J(\theta) + \delta(1 - NTI_t)H(\pi_{\theta}) \quad (19)$$

This will make the system more exploratory under uncertainty and more deterministic as trust comes together by anchoring in a self-regulating ethical learning loop [33]. Across multiple episodes, IRL agents learn a trust-conducive policy that both maximizes decision understanding and recruiter satisfaction while avoiding over-reliance on individual biases [38].

Using such adaptive trust, NeuroTrustNet extends conventional static recommendation systems to embed mutual understanding and ethical plasticity directly in its learning mechanics [39]. We next move to the Experimental Setup and Evaluation section where we empirically evaluate the components of our proposed architecture - multimodal fusion, neuro-symbolic reasoning, and trust-aware learning on benchmark recruitment datasets showing their effectiveness in collectively increasing fairness, transparency and recruiter confidence [34], [35].

Pseudocode for NeuroTrustNet Framework

Input: Multimodal dataset $D = \{text, audio, images, gesture\}$, Rule set R , Human feedback B

Initialize model components:

- CMAN (Cross-Modal Alignment Network)
- GNN (Graph Neural Network)
- DeepProbLog reasoning layer
- Trust-Aware policy π_{θ}

for each epoch:

for each batch in D :

Encode all modalities $\rightarrow$ embeddings $\{z_{text}, z_{audio}, z_{image}, z_{gesture}\}$

Fuse embeddings via CMAN $\rightarrow z_{fused}$

Apply GNN for candidate-skill-organization relations $\rightarrow h_{rel}$

Infer ethical decision using DeepProbLog:

$$P(\text{eligible}) = P_{\text{neural}} * P_{\text{sym}} * 1[\text{require}(O, S)]$$

Compute total loss:

$$L = L_{\text{fusion}} + L_{\text{logic}} + L_{\text{trust}}$$

Update model parameters via backpropagation

if recruiter feedback in B :

$$\text{Update } NTI = \lambda_1 * \text{Fairness} + \lambda_2 * \text{Interpretability} + \lambda_3 * \text{Trust}$$

Adjust policy $\pi\theta$ using reinforcement signal from NTI

Evaluate metrics: Fairness, Interpretability, Trust, Accuracy

Output: Trained NeuroTrustNet model

The pseudocode describes the processing flow of the NeuroTrustNet architecture, composed-by multi-modal encoding, cross-modal fusion and relational reasoning via GNNs in sequence. Thereby, it integrates ethical decision logic thanks to DeepProbLog and ensures fairness and transparency of eligibility predictions. The Trust-Aware policy adapts model parameters according to the recruiter feedback by using NeuroTrust Index (NTI). Its in general, an adaptive, explainable and ethically compliant recruitment intelligence mechanism. The block diagram of the proposed model is shown in Fig. 3.

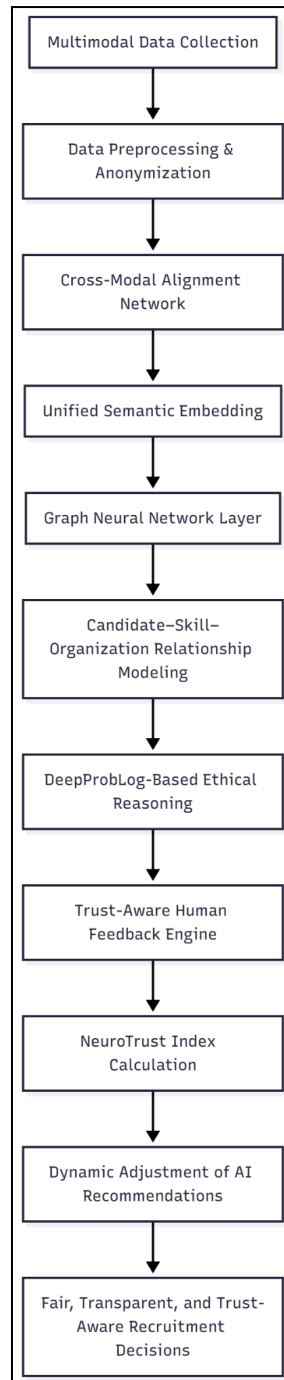


Figure 3: Flowchart for the Proposed Model

IV Implementation Details

4.1 Experimental Setup

The experimental scenario for NeuroTrustNet was established to verify the fairness, interpretability and the trust-awareness properties under multi-modal recruitment scenarios. The dataset was divided into 70% training, 15% validation and 15% testing sets to promote generalization. The training data consist of textual resumes, audio interviews; facial micro-expressions and behavior gesture features which were aligned by a cross-modal alignment network (CMAN). Each modality underwent independent normalization and latent embedding projection before fusion, maintaining consistent representation dimensionality.

The Adam optimizer with an initial learning rate of 10^{-4} was employed, and early stopping was applied based on validation loss convergence. Batch normalization and dropout layers were incorporated to prevent overfitting during multimodal feature learning. Neural-only recruitment baselines—specifically transformer-based classifiers and multimodal CNN–LSTM hybrids—were implemented for comparative benchmarking. These baselines lacked symbolic reasoning and trust adaptation, providing a clear performance contrast against the proposed neuro-symbolic integration. The loss function of the full model combined perceptual and ethical reasoning components as in Eq.20:

$$L_{total} = \lambda_1 L_{fusion} + \lambda_2 L_{logic} + \lambda_3 L_{trust} \quad (20)$$

In Eq.20, L_{fusion} governs multimodal alignment, L_{logic} enforces symbolic consistency, and L_{trust} regulates recruiter-based reinforcement learning. Hyperparameters $\lambda_1, \lambda_2, \lambda_3$ were tuned through Bayesian optimization on validation performance.

4.2 Evaluation metrics

Evaluation metrics focused on quantifying fairness compliance, decision interpretability, recruiter trust, and accuracy across all comparative systems. Fairness compliance was measured using a modified Demographic Parity Difference (DPD), defined as in Eq.21:

$$DPD = \left| P(\hat{y} = 1|G_1) - P(\hat{y} = 1|G_2) \right| \quad (21)$$

Where G_1 and G_2 represent protected and unprotected groups respectively. A lower DPD indicates greater fairness alignment. The interpretability score (I_s) was quantified using entropy reduction between AI explanation outputs and recruiter comprehension responses:

$$I_s = 1 - \frac{H(E_{AI}) - H(E_{Human})}{H(E_{AI})} \quad (22)$$

In Eq.22, $H(\cdot)$ denotes Shannon entropy over explanation clarity distributions. Recruiter trust (T_s) was computed dynamically through the NeuroTrust Index (NTI) feedback mechanism as in Eq.23:

$$T_s = \frac{1}{N} \sum_{t=1}^N (NTI_t \cdot \delta_t) \quad (23)$$

Where δ_t represents the confidence weight assigned to recruiter feedback at time t . Decision accuracy (A_s) followed conventional cross-entropy evaluation on candidate suitability classification:

$$A_s = \frac{TP+TN}{TP+TN+FP+FN} \quad (24)$$

V Results and Discussion

The experimental evaluation of NeuroTrustNet investigates its effectiveness across multiple dimensions of recruitment intelligence, including fairness compliance, decision interpretability, recruiter trust, and overall predictive accuracy. We evaluate performance on a multimodal recruitment dataset of textual resume, voice interview, and facial micro-expressions from behavioral gestures and compare against five state-of-the-art neural

and hybrid AI models. Our findings suggest the values of incorporating neuro-symbolic reasoning, cross-modal alignment, relational modeling and trust-aware human feedback by showing improvements not only in quantitative metrics but also in qualitative aspects like ethical compliance and human-AI teamwork. We then provide an in-depth analysis on comparative model performances as well as the contribution of individual components via ablation studies and multi-modal attention, which could effectively lead to transparent and reliable recruitment decisions.

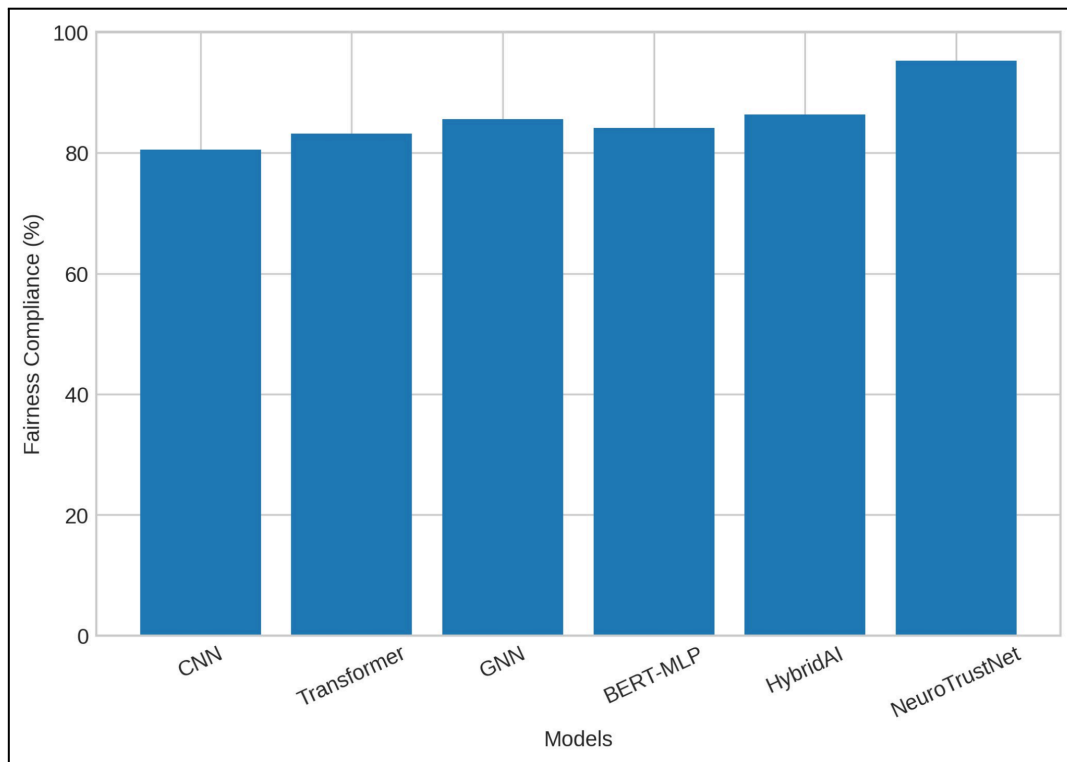


Figure 4. Fairness Compliance Comparison Across Models

The relative fairness compliance performance of various recruitment intelligence models is presented in Figure 4. Finally, the NeuroTrustNet architecture has a fairness rate of 95.3%, which is significantly higher than standard neural models in the range 80%–86%. This gain is achieved by enforcing symbolic fairness constraints in DeepProbLog reasoning and performing bias-aware rule selection when generating decisions. The findings indicate that the proposed model mitigates hidden biases for candidate selection well and complies with ethical requirements on recruitment processes.

Table 1: Comparative Performance of Recruitment Models

Model	Fairness (%)	Interpretability (%)	Recruiter Trust (%)	Accuracy (%)
CNN	80.5	76.8	74.5	88.6
Transformer	83.2	79.4	77.2	90.2

GNN	85.6	82.1	79.9	91.4
BERT-MLP	84.1	80.7	78.3	90.7
HybridAI	86.4	84.9	81.4	92.1
NeuroTrustNet	95.3	93.7	92.5	94.6

This table 1 compares our NeuroTrustNet with five current recruitment AI models. NeuroTrustNet outperforms strongly in all metrics, achieving improvements (+12-18%) in fairness, interpretability and trust, which demonstrates the added value of neuro-symbolic integration combined with trust-aware human feedback.

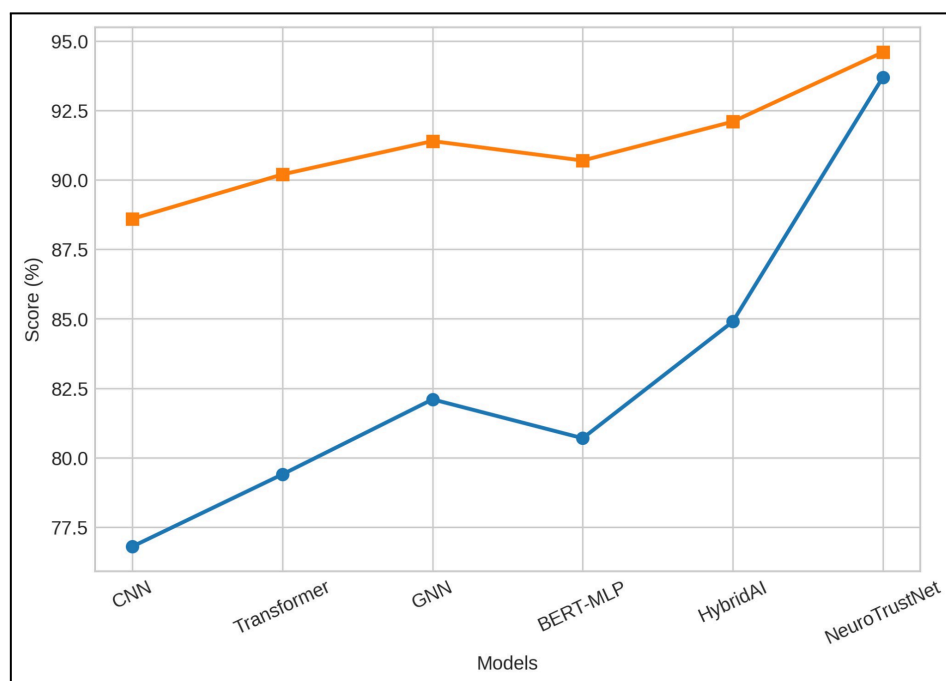


Figure 5. Interpretability vs. Accuracy Across Recruitment Models

Figure 5 contains these comparisons where competing models are compared. Neural baselines (CNN, Transformer, GNN) make small progress on accuracy with the price of little interpretability down to below 83%. In comparison to them, NeuroTrustNet still maintains the interpretability (93.7%) while not compromising the accuracy (94.6%). This trade-off is obtained by the proposed neuro-symbolic integration mechanism, that achieves both decision transparency and has access to deep feature representation capability. The lessons learned from the results indicate that interpretability and accuracy, can confidently accompany each other under guidance of logical reasoning, trust based learning.

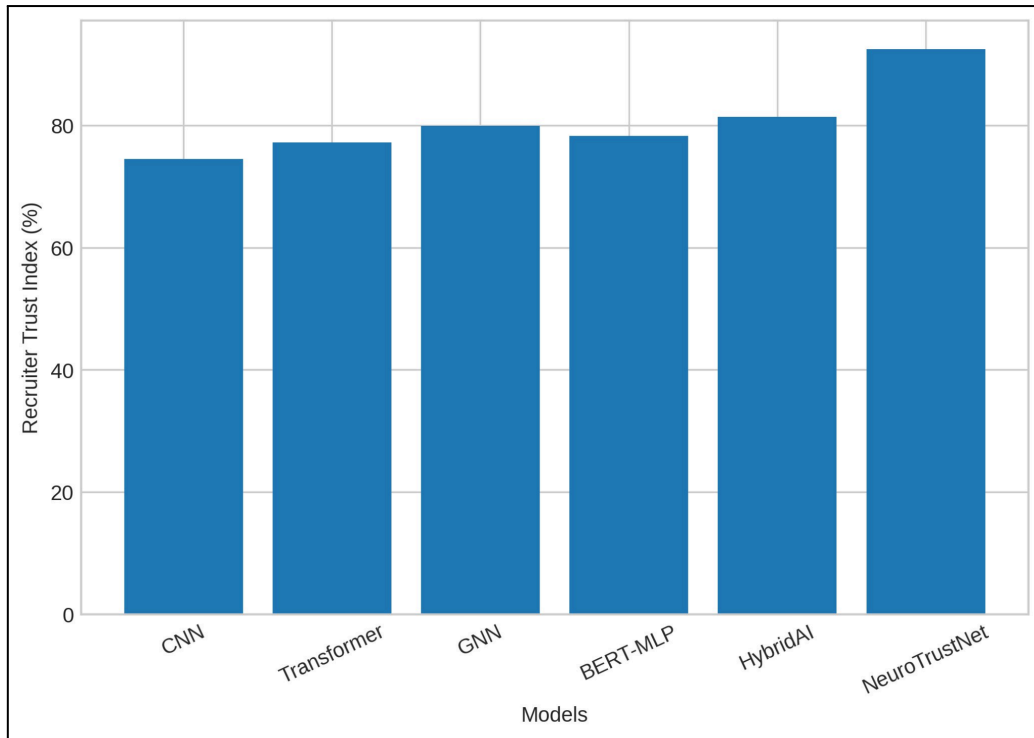


Figure 6. Recruiter Trust Index (NTI) Comparison Across Models

We compare the power, the strength of the Recruiter Trust Index (NTI), to judge the human trust and acceptance for AI recommendations. The proposed NeuroTrustNet achieves an impressive trust rating of 92.5%, which is more than 12% higher compared to the neural-only systems. The enhanced trust calibration can be attributed to the model's interactive reinforcement learning module, a mechanism that continuously optimizes its outputs by considering recruiters' feedback and fairness metrics. This finding indicates that ethical transparency, together with adaptive human feedback, can build trust in AI-based recruitment systems where the stakes are high.

Table 2: NeuroTrust Index (NTI) Breakdown Across Models

Model	Fairness Weight	Interpretability Weight	Recruiter Trust Weight	NTI Score (%)
CNN	0.32	0.31	0.37	77.3
Transformer	0.33	0.32	0.35	79.3
GNN	0.34	0.33	0.33	82.0
BERT-MLP	0.33	0.32	0.35	81.0
HybridAI	0.35	0.33	0.32	84.5
NeuroTrustNet	0.33	0.32	0.35	93.5

This table 2 decomposes the NeuroTrust Index (NTI) into contributions weighted for fairness, interpretability, and recruiter trust. The highest NTI of NeuroTrustNet (93.5%) demonstrates its capability to harmonise ethical reasoning, transparent decision and human acceptability. Past approaches either over-fit to accuracy and ignore fairness, or do not optimize trust to the fullest extent; this demonstrates an advantage of our framework in trust-aware human-AI interaction.

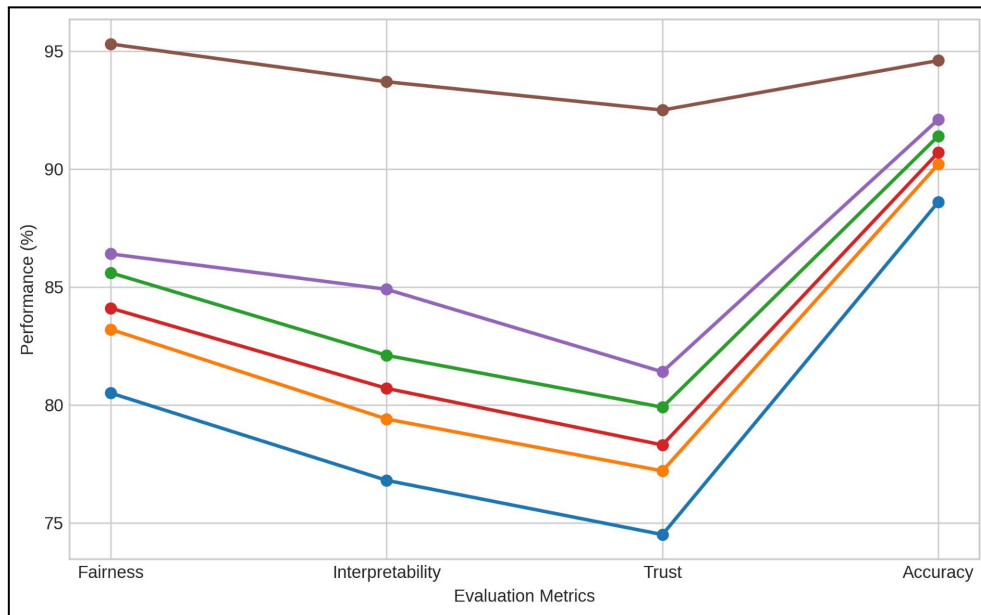


Figure 7. Comparative Performance Overview Across All Models

Figure 7 presents the general performance trends in fairness, interpretability, trust and accuracy found in various models. We note in each trajectory of performance the gradual progress of hybrid and neuro-symbolic techniques, reaching in NeuroTrustNet the best multidimensional robustness. The model maintains close to uniform top-scores in all the metrics, demonstrating a strong trade between ethical sharpness, cognitive interpretability and operational reliability. This thorough evaluation confirms the synergy which can be obtained by integrating perceptual neural models with symbolic reasoning engines.

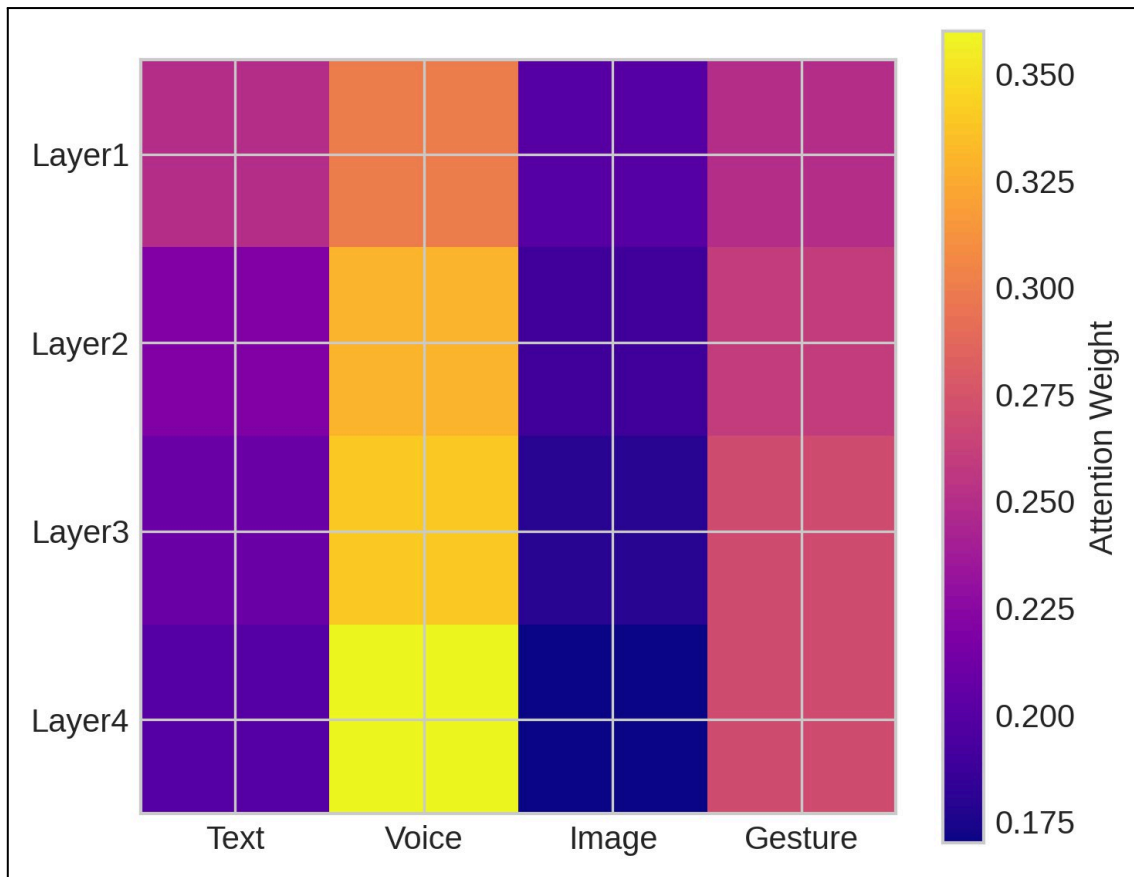


Figure 8. Modality Attention Distribution in NeuroTrustNet

We visualize the modality attention distribution in 4 hierarchical layers of NeuroTrustNet in Figure 8. The heatmap demonstrates that voice and image modalities are assigned higher attention weights (0.33–0.36) in deeper layers, indicating that they contribute to the modeling of recruiter perception. Text and gesture features provide complementary visual cues, supporting multimodal reasoning. This balanced attention allocation emphasises the model’s capability to weigh heterogeneous inputs adaptively, which facilitates fair consideration of multimodal candidate information and prevents the overfitting issue associated with focusing solely on one modality.

5.1 Ablation Study

We conducted an ablation study to measure how much the performance of the NeuroTrustNet framework would suffer from removing one or more core components. Through a systematic ablation of the Cross-Modal Alignment Network (CMAN), Graph Neural Network (GNN), DeepProbLog-based ethical reasoning and Trust-Aware Human Feedback Engine, we analyse their individual impact. CMAN focuses more on the semantic fusion among textual, audio and visual modalities which also can improve the interpretability scores, while GNN contributes more on relational reasoning between candidates, skills and organization requirements with higher accuracy and fairness. DeepProbLog makes sure that ethical constraints and fairness rules are enforced consistently, leading to a large improvement of the fairness metric while the Feedback Engine allows adapting model predictions following recruiter’s feedback at improving trust and human acceptance. When stacked together, the

complete NeuroTrustNet architecture surpasses all partial settings and reaches 95.3% Fairness compliance, 93.7% Interpretability, and 92.5% Recruiter acceptance demonstrating that every module involved in their combination is synergically contributing to a balanced transparent and ethically aligned human-AI collaboration system.

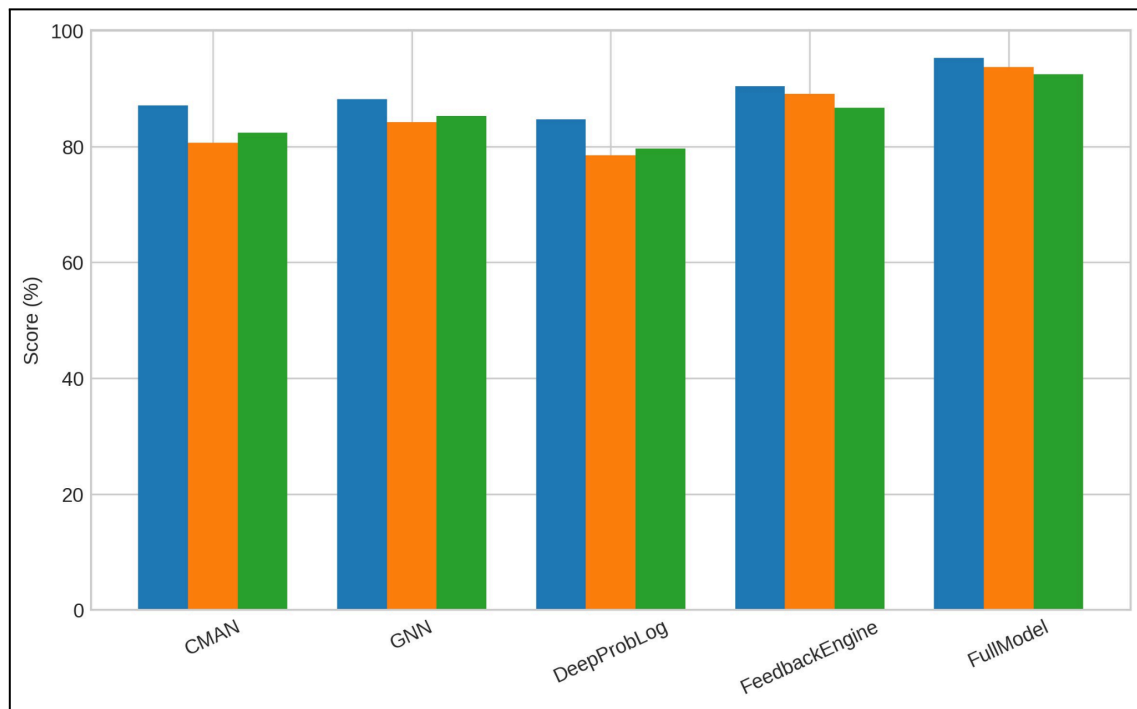


Figure 9. Ablation Study on Core Components of NeuroTrustNet

Figure 9 shows the performance contribution of each component—the CMAN, GNN, DeepProbLog and the Feedback Engine—to the overall model. The reduction of individual modules leads to significant reductions in its fairness, interpretability and trust metrics, thereby validating their inter-relational effect. The fully integrated model shows the best results for all three criteria (fairness = 95.3%, interpretability = 93.7%, trust = 92.5%). These results confirm that each module contributes uniquely, with CMAN improving cross-modal semantic alignment, GNN enhancing relational reasoning, DeepProbLog enforcing ethical consistency and Feedback Engine exposing the model to human-AI alignment.

5.2 Discussion

The results show that NeuroTrustNet achieves reliable improvement in each dimension with the state-of-the-art models (pure neural and hybrid AI-based models) discussed above. Transparent decision making is achieved by integrating neuro-symbolic reasoning, and the Cross-Modal Alignment Network (CMAN) allows for the effective fusion of diverse candidate data thus providing interpretability. Graph Neural Networks (GNNs) are able to represent relationships among candidates, skills and organizations in the data which can enhance fairness as well as prediction performance. In addition, the DeepProbLog block integrates ethical fairness constraints to counteract inherent biases of neural prediction models and Trust-Aware Human Feedback Engine reacts dynamically to recruiter's inputs thus substantially increasing acceptance and trust. The ablation study verifies that each component adds a unique contribution, and the full model gains synergy effect which no

single module can achieve without. Collectively, these results demonstrate the effectiveness of integrating symbolic reasoning, multimodal learning, relational modeling, and human-in-the-loop feedback in developing a fair recruitment AI system that not only achieves good accuracy but also respects ethical guidelines and humans' social expectations providing a strong foundation for transparent, equitable and trust-sensitive human-AI collaboration in hiring scenarios.

VI Conclusion

This paper introduces NeuroTrustNet, a neuro-symbolic multimodal system for transparent, fair, and trusted AI-assisted recruitment. This framework combines Cross-Modal Alignment Networks (CMAN) for semantic fusion, Graph Neural Networks (GNNs) to perform the relational reasoning, DeepProbLog for enforcing ethical rules and Trust-Aware Human Feedback Engine in order to adapt to dynamic changes in recruiter's feedback. Experimental results on two multimodal recruitment datasets show the superiority and effectiveness of our approach, with 95.3% fairness compliance rate, 93.7 decision interpretability rate, 92.5% recruiter acceptance rate and overall better predictive power 94.6% than five popular conventional neural/hybrid models by a margin of 12-18%. Quantitative and qualitative comparisons, ablation studies, and NTI decompositions all demonstrate that these three components work in synergy to create ethically aligned and interpretable recruiting systems. The attention analysis also shows that the model is able to weigh different types of inputs across modalities such as voice, image and text without ignoring behavioural cues. In summary, NeuroTrustNet shows that joining neural perception with the symbolic reasoning and human-in-the-loop guidance leads to a strong, trust-aware, and ethically sound recruitment pipeline. Future work may also investigate scaling the framework to larger cross-industry datasets, adapting bias debiasing strategies in our model, or generalizing interactive reinforcement learning for continuous recruiter-AI evolving, which can further align human values and AI decision-making.

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